Optimization and Calibration of STELLA: Lab and Field Methods for Local Spectroscopy Measurements

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STELLA: An Introduction

The STELLA is a handheld spectrometer that measures several wavelengths of light, humidity, pressure, and temperature (both ambient and surface) to assess overall plant health. STELLA provides ground-level measurements that can be overlaid with LandSAT imagery for detailed comparison and calibration. "Both Landsat and STELLA sensors fundamentally rely on the same physics principles - they detect electromagnetic radiation across defined wavelength ranges using photodetectors." [1]

Figure 1: Image of Operational Land Imager for LandSAT 8

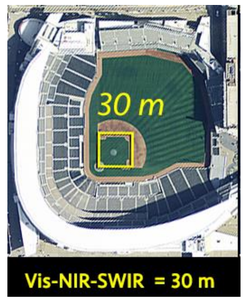


Figure 2: Spatial Resolution of LandSAT 8 OLI

Motivating Questions:

- How can our team validate the STELLA for environmental research purposes through both laboratory and field testing?
- What Procedures will properly test the STELLA?
- Why is a field protocol necessary? To improve consistency of data collection and usability of data.
- What is a proper field protocol?
- How accurate are the STELLA's sensors and can we compensate inaccuracies?

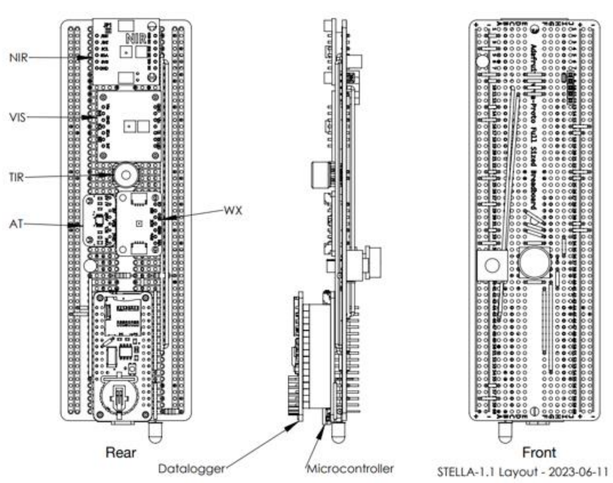
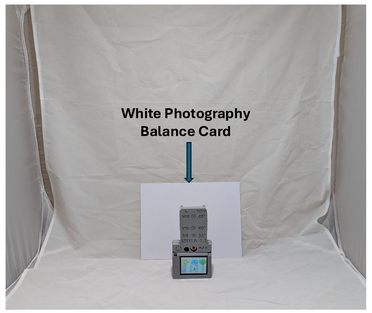
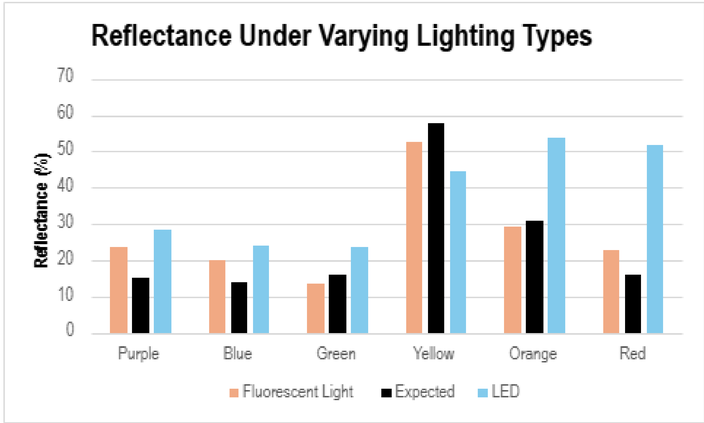


Figure 3: Diagram of STELLA 1.1

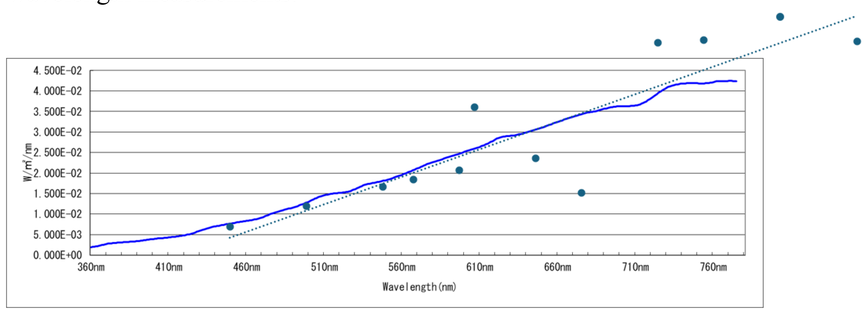
Experimentation:

- We set up a Light Box to isolate from outside light influence
- Two 6500K, 50W LED Light Bulbs were used to match the color temperature the sensors were originally calibrated in
- Yielded different results than testing under fluorescent lightbulbs or direct sunlight





The subjects of the experiments were paint cards with manufacturer-provided reflectance values. Our data points were averaged across multiple visual light wavelength measurements.



Graph showing readings taken with the STELLA 1.1 and a laboratory-grade spectral radiometer* of the same halogen lamp compared with each other.

Blue Line: Lab-Grade Spectral Radiometer
Grey Points: STELLA Data
Dotted Line: Line of Best Fit for STELLA data

*All readings taken at the University of Colorado Boulder Lighting Laboratory

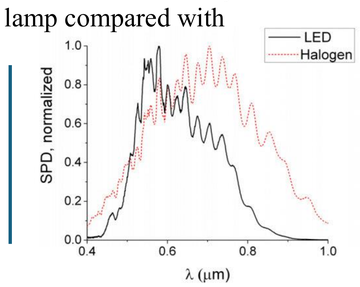


Figure 4: Spectral Power Distribution of an LED & Halogen Lamp

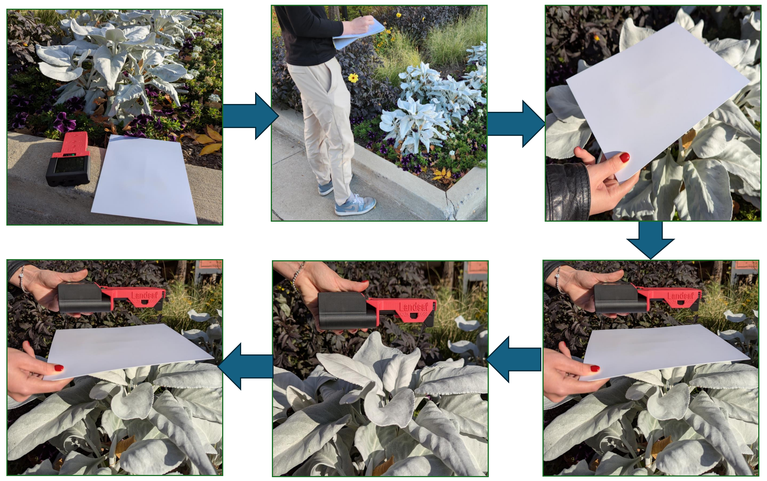
Conclusions:

The "Calibration Sandwich" Method of data collection is detailed below.

Step 1: Identify the Subject and prepare the Reference Material

Step 2: Record the Conditions (Ideally uniform sunlight)

Step 3: Place the Reference Material close to Subject



Step 4: Take measurement of the Reference Material (From Approx. 10cm away)

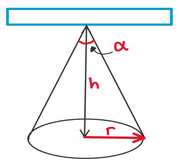
Step 5: Quickly remove Reference Material and take a measurement of the Subject

Step 6: Repeat the Reference Material measurement, note any differences with the first measurement.

How changing the Angle of Incidence changes the Area measured:

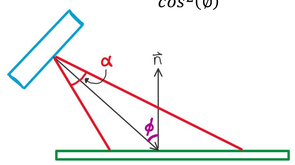
Perpendicular to Surface:

$$Area = \pi r^2 = \pi h^2 tan^2(\frac{\alpha}{2})$$



Angled to Surface:

$$A(\emptyset) = \frac{\pi h^2 tan^2(\alpha/2)}{cos^2(\emptyset)}$$



Future Research

- A STELLA 1.2 device is in development with new sensors and will be tested with the same procedures
- Continue to develop testing procedures and field protocols
- Create more educational content (videos and guides)

References:

[1] Michael P. Taylor, Personal Communication, July 17th, 2025
Fig 1) NASA LandSAT Website
Fig 2) NASA LandSAT Website
Fig 3) NASA STELLA 1.1 Build Instructions
Fig 4) OpticsExpress LED Optical Coherence Microscopy